

```
1.import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame(np.arange(0,20).reshape(5,4),
                  index=['Row1','Row2','Row3','Row4','Row5'],
                  columns=["Column1","Column2","Column3","Column4"])

df.head()    # Displays top 5 rows
df           # Displays entire DataFrame

# Accessing data using loc (label based)
df.loc['Row1']          # Series
type(df.loc['Row1'])    # Output: <class 'pandas.core.series.Series'>

# Accessing column
df['Column1']           # Series
type(df['Column1'])    # Output: <class 'pandas.core.series.Series'>

# Accessing using iloc (position based)
df.iloc[0:2, 0:2]       # Returns 2x2 DataFrame
type(df.iloc[0:2, 0])   # Series

# Check null values
df.isnull()
```

```
df.isnull().sum()
```

```
# Unique and value counts
```

```
df['Column1'].unique()
```

```
df['Column1'].value_counts()
```

```
# Select multiple columns
```

```
df[['Column2','Column3']]
```

```
# Convert to CSV
```

```
df.to_csv("Test1.csv")
```

```
# Convert to NumPy array
```

```
df.iloc[:,:].values
```

```
df.iloc[:,:].values.shape    # Output: (5, 4)
```

```
2. import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV
```

```
df = pd.read_csv(r'C:\Anuradha\gapminder-FiveYearData.csv')
```

```
df.head()    # First 5 rows
```

```
df.tail()    # Last 5 rows
```

```
df.shape      # (rows, columns)
df.columns    # Column names
df.dtypes     # Data types
df.info()     # DataFrame info
```

```
# Access column
```

```
ctry = df['country']
```

```
ctry.head()
```

```
ctry.tail()
```

```
ctry.head(10)
```

```
ctry.tail(10)
```

```
# Access multiple columns
```

```
subset = df[['country','continent','year']]
```

```
# loc and iloc
```

```
df.loc[0]
```

```
df.loc[[0,99,999]]
```

```
df.iloc[1]
```

```
df.iloc[-1]    # Last row
```

```
# Subset rows & columns
```

```
df.loc[0:99, ['country','year']]
```

```
df.iloc[:, [1,3,5]]
```

Filtering

```
df[df.lifeExp >= 60]
```

```
df.loc[(df.lifeExp >= 60) & (df.country == 'Albania')]
```

```
df.loc[(df.lifeExp >= 60), ['country','pop','lifeExp']]
```

Update values

```
df.loc[df.country == 'Albania', ['pop']] = 100000
```

GroupBy operations

```
df.groupby('year')['lifeExp'].mean()
```

```
multi_gp =
```

```
df.groupby(['year','continent'])[['lifeExp','gdpPercap']].mean().reset_index()
```

Unique country count per continent

```
df.groupby('continent')['country'].nunique()
```

3. import pandas as pd

```
import numpy as np
```

Read CSV

```
df = pd.read_csv(r"D:\Anuradha\mercedesbenz.csv")
```

```
df.head()
```

```
df.info()
```

```
df.describe()
```

```
# Create DataFrame from np.arange
df1 = pd.DataFrame(np.arange(20).reshape(5,4),
                    index=["Row1","Row2","Row3","Row4","Row5"],
                    columns=["Column1","Column2","Column3","Column4"])
df1.describe()
df1.to_csv("Test1.csv")
df2 = pd.read_csv("Test1.csv")
df2["Column1"].unique()
df2["Column1"].value_counts()

# Custom separator
pd.read_csv("Test1.csv", sep=';')

# Conditional selection
df['y'] < 100
df[df['y'] < 100]
df.corr()

# Handling NaN
data = [[1,2,3],[4,5,np.nan],[6,7,np.nan],[np.nan,np.nan,np.nan]]
df3 = pd.DataFrame(data)
df3.dropna()          # Drop all NaNs
df3.fillna("Missing") # Replace NaNs

# Random values
```

```
df4 = pd.DataFrame(np.random.randn(5,3))
```

```
# Reindexing
```

```
df5 = df4.reindex(['a','b','c','d','e'])
```

```
df5
```

```
# Dictionary to DataFrame and reindex
```

```
df =
```

```
pd.DataFrame({"A":[1,5,3,4,2],"B":[3,2,4,3,4],"C":[2,2,7,3,4],"D":[4,3,  
6,12,7]}), index=["first","second","third","fourth","fifth"])
```

```
df.reindex(["first","dues","trois","fourth","fifth"], fill_value=100)
```

```
4. import pandas as pd
```

```
from io import StringIO
```

```
# Basic CSV data
```

```
data = 'col1,col2,col3\nx,y,1\na,b,2\nc,d,3'
```

```
pd.read_csv(StringIO(data)) # Full Data
```

```
pd.read_csv(StringIO(data), usecols=['col1','col3'])
```

```
# Missing value handling
```

```
data1 = 'a,b,c,d\n1,2,3,4\n5,6,7,8\n9,10,11' # Missing one value
```

```
df = pd.read_csv(StringIO(data1))
```

```
df.info()
```

```
# Custom dtype
```

```
df = pd.read_csv(StringIO(data1), dtype=object)
df.dtypes
```

```
# Index column usage
```

```
data2 = 'index,a,b,c\n4,apple,bat,5.7\n8,orange,cow,10'
pd.read_csv(StringIO(data2), index_col=0)
```

```
# Escape characters
```

```
data3 = 'a,b\n"hello, \\"Bob\\",nice to see you",5'
pd.read_csv(StringIO(data3), escapechar=\\')
```

```
5. import pandas as pd
```

```
# JSON (column oriented)
```

```
data = '{"emp_id": {"Sree": "101"}, "email": {"Sree":  
"sree@gmail.com"}, "position": {"Sree": "Team Lead"}}'  
pd.read_json(data)
```

```
# JSON with records
```

```
json_data = "[  
  { \"id\": \"101\", \"Name\": \"Emily\", \"Age\":22, \"Department\": \"HR\" },  
  { \"id\": \"102\", \"Name\": \"Charles\", \"Age\":22, \"Department\": \"IT\" }  
]"
pd.read_json(json_data, orient='records')
```

JSON with split

```
json_data = "{  
    "index":["Employee_1","Employee_2"],  
    "columns":["Id","Name","Age","Department"],  
    "data":[["101","Sree","32","Development"],  
            ["102","Santhosh","45","Testing"]]  
}"
```

pd.read_json(json_data, orient='split')

Reading HTML

```
url = 'https://en.wikipedia.org/wiki/Mobile_country_code'  
df_list = pd.read_html(url, match='Mobile network codes', header=0)  
df_list[0]    # First matching table
```

Excel

```
df = pd.read_excel('D:\Anuradha\Staff_Details.xlsx')  
df.head()
```

Convert CSV to JSON and vice versa

```
df.to_json(orient='records')  
df.to_json(orient='split')  
df.to_csv("output.csv")
```